onds, microseconds) by treating the local time offset as 0 and finding the UTC (year, month, day, hours, minutes, seconds, microseconds) tuple that corresponds to an elapsed time since the epoch time equal to the given value. The value then represents that tuple, but interpreted in the specific timezone and DST situation associated with the value. This procedure does not require knowing the local time offset of the value.

2. The value can be converted to a UTC time by subtracting the associated local time offset from the Epoch Time value and then treating the resulting value as an elapsed count of microseconds since the epoch time.

For example, an Epoch Time value of 0x0000_0BF1_B7E1_0000 corresponds to an offset of exactly 152 days. This can be interpreted as "00:00:00 on June 1, 2000" in whatever local time zone is associated with the value. That corresponds to the following times in ISO 8601 notation:

- 2000-06-01T00:00Z if the associated local time offset is 0 (i.e. the value is in UTC).
- 2000-05-31T23:00Z if the associated local time offset is +1 hour (e.g. the CET timezone, without daylight savings).
- 2000-06-01T00:00+02 if the associated local time offset is +1 hour.
- 2000-06-01T04:00Z if the associated local time offset is -4 hours (e.g. the EDT time zone, which includes daylight savings).
- 2000-06-01T00:00-04 if the associated local time offset is -4 hours.

Conversion from NTP timestamps

Timestamps from NTP also do not count leap seconds, but have a different epoch. NTP 128-bit timestamps consist of a 64-bit seconds portion (NTP(s)) and a 64-bit fractional seconds portion (NTP(frac)). NTP(s) at 00:00:00 can be calculated from the Modified Julian Day (MJD) as follows:

NTP(s) = (MJD-15020) * (24*60*60)

where 15020 is the MJD on January 1, 1900 (the NTP epoch)

NTP(s) on January 1, 2000 00:00:00 UTC (MJD = 51544) is 3155673600 (0xBC17C200)

Epoch Time has a microsecond precision, and this precision can be achieved by using the most significant 32 bits of the fractional portion (NTP(frac32)). Conversion between the 128-bit NTP timestamps and a UTC Epoch Time in Microseconds is as follows:

UTC Epoch Time = (NTP(s) - 0xBC17C200)*10^6 + ((NTP(frac32)*10^6) / 2^32) where all numbers are treated as unsigned 64-bit integers and the division is integer division.

7.19.2.4. Epoch Time in Seconds Type

This type has the same semantics as Epoch Time in Microseconds, except that:

- the value encodes an offset in seconds, rather than microseconds;
- the value is encoded as an unsigned 32-bit scalar, rather than 64-bit.

This type is employed where compactness of representation is important and where the resolution

